

NITHYA MADALAKOTE PRAKASH

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German: A1 (actively improving) | English: C1

PROFESSIONAL SUMMARY

CS graduate (B.Tech, 7.62 GPA) with hands-on experience in machine learning, Python development, and end-to-end AI application development. Currently completing a Data Science internship involving NLP, time series forecasting, and cloud deployment. Focused on building practical LLM, RAG, and MLOps solutions. Seeking a Werkstudent AI Engineer role in Germany to apply and deepen these skills in a professional engineering environment.

TECHNICAL SKILLS

Languages: Python (primary), SQL, C++, JavaScript

ML / AI: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, Hugging Face Transformers

LLM / GenAI: OpenAI API, LangChain, RAG pipelines, vector databases, prompt engineering

Data: Pandas, NumPy, Matplotlib, Seaborn, MongoDB, EDA, feature engineering

Web / APIs: FastAPI, Spring Boot, REST APIs, HTML/CSS/JavaScript, BeautifulSoup

Tools: Git, AWS (basics), Docker (learning), Jupyter, VS Code

WORK EXPERIENCE

Java Full Stack Developer Trainee — Palle Technologies

Bengaluru, India | Jun 2024 – Jan 2025

- Completed a structured 6-month full stack development programme, gaining experience in Java, Spring Boot, MySQL, and responsive front-end development.
- Built and tested REST APIs and integrated front-end and back-end components for internal web applications.
- Demonstrated strong problem-solving, code review participation, and team collaboration across the SDLC.

PROJECTS

Manufacturing Predictive Maintenance AI

Stack: Python, PyTorch, Scikit-Learn, Random Forest, Isolation Forest, SHAP, FastAPI, MLflow, Optuna, Streamlit, PostgreSQL, Redis, Docker Compose

- Built an end-to-end machine learning platform to forecast equipment failure and Remaining Useful Life (RUL), achieving 98% classification accuracy on simulated telemetry data.
- Implemented automated hyperparameter tuning (Optuna) and experiment tracking (MLflow) for Random Forest and Isolation Forest models.
- Deployed a microservices architecture using FastAPI, PostgreSQL, and Redis within Docker Compose, exposing an interactive Streamlit dashboard.
- Integrated SHAP values for Explainable AI (XAI), enabling automated root-cause analysis and actionable maintenance recommendations.

Transport Demand Prediction | Regression Modelling

Mar 2025 – May 2025 | Stack: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn

- Built a supervised regression model to predict intercity bus seat demand for Mobi Ticket (Kenya) using historical route, date, and traffic data.
- Engineered features including day-of-week indicators, weekend flags, and route encodings; trained and evaluated Linear Regression and Random Forest models using RMSE and R^2 .
- Conducted full EDA to surface demand patterns and booking trends influencing fleet allocation decisions.